

Linux GREP Cheatsheet

USAGE

```
$ grep [OPTION...] PATTERNS [FILE...]
$ grep [OPTION...] -e PATTERNS ... [FILE...]
$ grep [OPTION...] -f PATTERN_FILE ... [FILE...]
```

DESCRIPTION

grep searches each FILE for PATTERNS. PATTERNS is a string that contains one or more patterns separated by newline characters, and grep prints each line that matches a pattern. When using grep in a shell command, PATTERNS should usually be quoted.

A FILE of "-" denotes standard input. If no FILE is specified, recursive searches examine the working directory, while nonrecursive searches read standard input.

Furthermore, the variant programs egrep, fgrep, and rgrep are equivalent to **grep -E**, **grep -F**, and **grep -r**, respectively. These variants are obsolete, but remain available for backward compatibility.

OPTIONS EXAMPLES

`$ grep 'alias' filename` find all occurrences of a string (basic usage).

-c	<code>grep -c 'error' /var/log/syslog</code>	count the number of matches.
-e	<code>grep -e 'linux\$' filename</code>	use regex (lines ending with 'Linux').
-f	<code>grep -f pattern_file .bashrc</code>	take PATTERNS from FILE.
-i	<code>grep -i 'Alias' .bashrc</code>	ignore case search.
-l	<code>grep -l 'alias' /home/traw/*</code>	returns files with matches.
-L	<code>grep -L 'alias' /home/traw/*</code>	returns files without matches.
-r	<code>grep -r 'error' /var/log/log.txt</code>	recursive search (within subdirs).
-v	<code>grep -v 'warning' /home/traw/log.txt</code>	select non-matching lines.
-m	<code>grep -m 3 'alias' .bashrc</code>	limit grep Output upto 3 lines.
-n	<code>grep -n 'sudo' .bashrc</code>	print line numbers of the matches.
-n	<code>grep -n 'sudo' .bashrc</code>	print line numbers of the matches.
-o	<code>grep -o 'search string' .bashrc</code>	only show the matching part of the string.
-x	<code>grep -x 'linux is love' filename</code>	match only whole lines
-w	<code>grep -w "alias" .bashrc</code>	match only whole words
-A	<code>grep -A 6 'error' error.log</code>	print 6 lines after a match
-B	<code>grep -B 2 'warning' error.log</code>	print 2 lines before a match
-C	<code>grep -C 7 'info' error.log</code>	print 7 lines before and after a match
-E	<code>grep -E 'b(a e)g' filename</code>	Extended regex (lines containing bag or beg)

QUANTIFIERS

<code>{n}</code>	Matches exactly n times
<code>{n,}</code>	Matches n or More
<code>{,m}</code>	Matches up to m (maximum)
<code>{n,m}</code>	Matches between n and m (Minimum, Maximum)

WILDCARDS

<code>.</code>	Match any Character Except New Line
<code>?</code>	Match 0 or One characters
<code>*</code>	Match 0 or More characters
<code>+</code>	Match 1 or More characters

POSITIONS

<code>^</code>	Beginning of line.
<code>\$</code>	End of line.
<code>^\$</code>	Empty line.
<code>\<</code>	Start of word.
<code>\></code>	End of word.

CHARACTER CLASSES

<code>[A-Za-z]</code>	Any lower and upper case letter.
<code>[0-9]</code>	Any number.
<code>[0-9A-Za-z]</code>	Any lower and upper case letter or digit.



POSIX

<code>[:alpha:]</code>	Match any alphabetical character, either upper or lower case.
<code>[:alnum:]</code>	Matches any alphanumeric character 0–9, A–Z, or a–z.
<code>[:blank:]</code>	Matches a space or Tab character.
<code>[:digit:]</code>	Matches a numerical digit from 0 through 9.
<code>[:lower:]</code>	Matches any lowercase alphabetical character a–z.
<code>[:print:]</code>	Matches any printable character.
<code>[:punct:]</code>	Matches a punctuation character.
<code>[:upper:]</code>	Matches any uppercase alphabetical character A–Z.

BRE, ERE & PCRE

Basic Regular Expressions (BRE)

Unless they are preceded by a backslash, these characters have a special meaning in BRE.:

`^ $ . * [ ] \`

However, unless they are preceded by a backslash, these characters have no special meaning.:

`? + { } | ( )`

Extended Regular Expressions (ERE)

Unless they are escaped with a backslash, ERE gives each of these characters a unique meaning:

`^ $ . * + ? [ ] ( ) | { }`

Perl Compatible Regular Expressions (PCRE)

Additional anchors and character classes, lookahead/lookbehind, conditional expressions, comments, and other features are available in PCRE.

